

Esta é uma cópia de uma conversa entre ChatGPT e Anônimo.
Denunciar conteúdo

Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 rows** and **9 columns**, primarily focused on HIV epidemiological indicators among children and adolescents. Here's a summary of its key characteristics:

Columns:

- **Country/Region:** Geographic location.
- **UNICEF_Region:** Broader region classification; ~10,432 missing values.
- **Indicator:** Type of HIV-related metric.
- **Year:** Numeric year (as float).
- **Sex:** Gender category.
- **Age:** Age group.
- **Value:** The main metric value (currently stored as object type).
- **Lower/Upper:** Confidence interval bounds (some missing values, and stored as objects).

Observations:

- The dataset includes a `Value` column that may contain non-numeric data, requiring conversion to numeric for analysis.
- The dataset spans multiple indicators, age groups, and regions, making it suitable for detailed trend and comparison analyses.

Feel free to ask a specific analytical question you'd like answered using this data! [~]

Compare ARIMA and an LSTM-based deep learning model for predicting HIV incidence among adolescents (15–19) in the Americas (2024–2026). Evaluate performance using RMSE.

It looks like there's **no available data** in the dataset that matches all the filter conditions:

- **Region:** *Americas*
- **Age group:** *15–19*
- **Indicator:** *Incidence-related*
- **Sex:** *Total*
- **Year:** *Up to 2023* for training

This could be due to:

- Different labeling in the `Indicator` column
- Slight mismatches in the region or age categories

Let me first inspect unique values in the relevant columns (`UNICEF_Region`, `Age`, and `Indicator`) to understand the data structure better.

Thanks for your patience! Here's what I found:

Key Observations:

- The region you're looking for is labeled **"Latin America and Caribbean"**, not "Americas".
- The age group **"15–19"** exists, so that part is valid.
- Indicators related to incidence include:
 - **"Estimated incidence rate (new HIV infection per 1,000 uninfected population)"**
 - **"Estimated number of annual new HIV infections"** — This seems most appropriate for forecasting.

I'll now re-filter using:

- Region: **Latin America and Caribbean**
- Age: **15–19**
- Indicator: **Estimated number of annual new HIV infections**
- Sex: **Total**
- Use data up to 2023 for training and forecast 2024–2026

Let's prepare the dataset and preview it. [~]

Interesting—still no data coming through for that specific combination. Let's investigate a bit deeper:

I'll now **drop the `Sex == "Total"` condition** and check what years are available for:

- **"Estimated number of annual new HIV infections"**
- Region: **Latin America and Caribbean**
- Age group: **15–19**

This will help verify if the issue is with the `Sex` column or if data is simply missing. [~]

Perfect — now we've got data for the **15–19 age group** in **Latin America and the Caribbean** for **annual new HIV infections** from **2005 to 2023**.

We'll now:

1. **Train two models:**

- **ARIMA** (classic time series)
 - **LSTM** (deep learning)
2. Forecast for **2024–2026**
 3. Evaluate both using **Root Mean Square Error (RMSE)** over those forecasted years

Let's start by training the **ARIMA model** and generating its forecast. [\[↗\]](#)

ARIMA Forecast (Annual New HIV Infections, Age 15–19, Latin America & Caribbean):

- **2024:** ~1,800
- **2025:** ~2,084
- **2026:** ~1,877

Now let's train an **LSTM deep learning model** on the same data and generate its forecast for comparison.

[\[↗\]](#)

The environment doesn't currently have **TensorFlow** installed, so I can't run the LSTM model directly here.

However, I can still provide you with the complete Python code to run on your own machine with TensorFlow installed. Would you like me to give you that? It includes preprocessing, training, and forecasting using LSTM. [\[↗\]](#)

Envie uma mensagem para o Data Analyst



O ChatGPT pode cometer erros. Considere verificar informações importantes.